MOHAMMED BAJHAW

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Professional Summary

Robotics & Autonomous Systems Engineer with hands-on R&D experience in autonomous ground robots and UAVs, gained through TÜBİTAK 1001 and 2209-A funded projects at Fırat University's Robotics & AI Laboratory. Experienced across the full development cycle: hardware and sensor integration, embedded systems, 3D design and prototyping, simulation, field testing and maintenance. Builds reliable robotic systems with ROS 2, SLAM, LiDAR, thermal/RGB vision and Edge AI on NVIDIA Jetson and STM32 platforms. Co-author of three peer-reviewed publications.

Education

Bachelor of Science in Mechatronics Engineering | Fırat University, Elazığ, Türkiye *2021 – 08/2026*

Technical Skills

Robotics & Autonomy: ROS 1 / ROS 2, SLAM (LiDAR-inertial), Nav2, Path Planning, Obstacle Avoidance, Sensor Fusion, PID Control, Gazebo, RViz.

Embedded Systems & Electronics: NVIDIA Jetson (AGX Orin, Orin NX, TX2), STM32, ESP32, Raspberry Pi, Pixhawk, UART, CAN-Bus, LoRa, Wiring & Power Regulation, Circuit Board Maintenance.

Robots & Platforms: DJI Matrice 350 RTK (Payload SDK, E-Port), Unitree Go2, TurtleBot2 / TurtleBot3.

Sensors: 3D LiDAR (Velodyne VLP-16), Intel RealSense, Thermal Cameras (FLIR Lepton), FMCW Radar, IMU.

AI & Computer Vision: YOLO, TensorRT, OpenCV, PyTorch, Edge AI, Offline LLMs (llama.cpp), Speech AI (Faster-Whisper, Piper TTS).

Mechanical Design & Prototyping: SolidWorks, 3D CAD, 3D Printing, Rapid Prototyping, CNC / G-code, MATLAB / Simulink, Simscape Multibody.

Programming & Tools: C++, Python, Linux, Git / GitHub, VS Code, Jupyter.

Professional Experience

Robotics R&D (UAV & Ground Systems) | TÜBİTAK 1001 – RAI Lab, Fırat University *07/2025 – 08/2026*

- TÜBİTAK 1001 project "Next Level Autonomy Development for Fast and Robust Disaster Management using Heterogeneous Robots" (Grant No. 123E406) – responsible for the aerial-robotics (UAV) work package.
- Designed and integrated the sensor payload of a DJI Matrice 350 RTK (Jetson Orin NX, Velodyne VLP-16 LiDAR, RealSense D435i) via DJI E-Port and Payload SDK; streamed the drone's built-in IMU into ROS 2, removing the need for an external IMU.
- Deployed and tuned real-time LiDAR-inertial SLAM onboard the Jetson for GPS-denied mapping; evaluated SLAM methods (LIO-SAM, RKO-LIO) in Gazebo before field deployment.
- Fabricated custom 3D-printed parts (mounts, enclosures, brackets); assembled, wired, maintained and field-tested ground robots (Unitree Go2, TurtleBot) and drones.
- Collected and organized a 20,000+ camera and LiDAR image dataset for disaster-scenario human detection; co-authored 3 peer-reviewed papers.

Robotics R&D Project Lead | TÜBİTAK 2209-A – RAI Lab, Fırat University *02/2026 – 09/2026*

- Sole developer of an offline AI payload that turns any ROS 2 ground robot into a search-and-rescue robot: Jetson TX2 AI core + STM32 real-time sensor hub. Project page: bajhaw.com/projects/offline-multi-sensor-ai-payload-for-usar
- Integrated thermal (FLIR Lepton 3) + RGB cameras, a 24 GHz FMCW bio-radar, and LiDAR; deployed YOLO victim detection with TensorRT and a thermal "Wake-on-Suspect" power mode.
- Built a fully offline voice assistant (Faster-Whisper STT, Gemma 3 LLM via llama.cpp, Piper TTS) to guide trapped victims, with a SIGKILL-based state machine for safe AI-model switching on limited memory.
- Developed hazard-aware navigation (Nav2, Cartographer) and dual Wi-Fi + LoRa telemetry with a command-center receiver; validated on TurtleBot3 and TurtleBot2. Research paper under review.

Publications

- **MSR-SLAM: A Multi-Stage Map-Refinement Framework for Reliable LiDAR–Inertial SLAM in GPS-Denied Environments** Applied Sciences (MDPI), 2026 | A multi-stage refinement pipeline that improves the reliability of LiDAR-inertial SLAM in GPS-denied environments.
- **PV-STAM: Velocity-Aware Attention for Mapless Deep Reinforcement Learning Navigation in Dynamic Environments** Applied Sciences (MDPI), 2026, 16(18), 9083 | Obstacle-motion awareness from single-frame 2D LiDAR, validated in simulation and on a TurtleBot3.
- **Drift-Resilient Indoor 3D Mapping in GPS-Denied Environments Using RKO-LIO with Velodyne–Pixhawk Integration in ROS 2** IEEE ICHORA 2026 | High-precision indoor 3D mapping and localization through Velodyne LiDAR and Pixhawk integration in ROS 2.
- **Under Review:** Demonstrating the Efficiency of RKO-LIO on a DJI UAV Platform Utilizing Drone Built-in IMU for GPS-Denied Mapping.
- **Under Review:** Offline Multi-Sensor AI Payload for Autonomous Victim Detection and Voice Interaction in Search and Rescue Robots.

Projects

5-Axis Robotic Arm Design & Simscape Multibody Control *01/2026*

- Designed a 5-axis robotic arm in SolidWorks and imported it into MATLAB/Simulink Simscape Multibody for physics-based simulation.
- Implemented inverse kinematics (geometric method, atan2, law of cosines) to convert XYZ targets into joint angles; studied PID control on a 2-DOF model. GitHub: github.com/MohammedBajhaw/Robotic-Arm-SolidWorks-Matlab

OA-Drone – Autonomous Cave Exploration & Obstacle Avoidance (ROS 2) *11/2025*

- Built a ROS 2 Humble + Ignition Gazebo simulation of an Iris quadrotor exploring GPS-denied caves, tunnels and mines, creating 3D maps with OctoMap and collision-free path planning with D*+/DSP planner.
- Developed a Python flight controller with look-ahead path tracking, altitude control and stuck/collision recovery. GitHub: github.com/MohammedBajhaw/drone-navigation

Automated Coffee Preparation System *06/2025*

- Engineered an automated beverage preparation machine with Raspberry Pi running modular OOP Python to coordinate mechanical and electrical operations.
- Integrated relay-switched high-power loads (220V boiler, 12V pump), PWM-driven dispensing servos, a 16x2 LCD/matrix keypad UI, and SSH over Ethernet.

Multi-Function CNC Conversion Machine *03/2025*

- Converted a 3D printer into a multi-functional 3-axis CNC router with interchangeable tool heads (drill, heated pen, writing pen) for carving, engraving, and drawing.
- Designed mechanical modifications in SolidWorks, calibrated lead screw backlash and stepper motion, and built a CAD/CAM toolpath pipeline (Inkscape, JSCut, Cura) for G-code execution.

Certifications & Training

Commercial Unmanned Aerial Vehicle Pilot License (UAV-1)

Drone-Based Terrain Survey and Mapping Training – Elazığ, 2025

Udemy: ROS II (Localization, Navigation, SLAM)

Coursera: Introduction to Automotive Embedded Systems (University of Colorado Boulder)

DeepLearning.AI: Supervised Machine Learning | Advanced Learning Algorithms

Languages

Arabic (Native) | Turkish (C1) | English (B2)